

Reversing in-vitro adaptations

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1 Abstract

In a recent perspective we proposed that tissue microenvironments that lack sufficient resources to support tetraploid cells can explain the fitness cost of aneuploidy. In this perspective we hypothesized that both Oxygen and Phosphate, but not Carbon are Rate-Limiting Factors for High-ploidy Cells. Here we test this hypotheses by exposing cancer cell lines to long-term evolution under either limiting Glucose, Phosphate or Oxygen, finding that ...

Alternative: Karyotype heterogeneity is widespread in cancers, yet its functional advantages in tumor adaptation remain poorly understood. Here, we identify that karyotypic diversity confers a survival benefit by enabling cancer cells to adapt to diverse microenvironments, including glucose deprivation, phosphate limitation, and hypoxia. Using g-banded karyotyping, microscopy-based cell imaging and unbiased metabolomics, we demonstrate that heterogeneous karyotypes facilitate metabolic flexibility, allowing subsets of cells to thrive under nutrient-deprived conditions. High-ploidy variants exhibit enhanced survival in glucose-starved environments, while specific aneuploid clones dominate under phosphate and oxygen scarcity (will be assessed soon?). Overall, our findings indicate that karyotype heterogeneity serves as an evolutionary reservoir, promoting clonal selection and tumor persistence in fluctuating conditions. Our study provides critical insight into how genomic instability drives cancer cell adaptation and suggests that targeting karyotypic plasticity could be a novel therapeutic strategy to disrupt tumor survival mechanisms across diverse microenvironments.

2 Introduction

Cancer cells sampled from tumor patients typically undergo considerable genomic and phenotypic evolution within just five passages of in-vitro growth.^{1,2} Standard in-vitro growth conditions differ significantly from those in-vivo. For example, RPMI medium contains elevated glucose levels at 11 mM—approximately three times higher than physiological concentrations—as well as threefold higher phosphate levels.³ Oxygen availability also differs drastically.

These differences lead to notable alterations in both cell phenotype and genotype. One of the most striking genotypic differences is in ploidy: while most cancer cell lines in-vitro are near-triploid, in-vivo cells tend to exhibit a near-diploid ploidy peak. This raises the question of whether key differences in resource availability—such as glucose, phosphate, oxygen, and space—are responsible for driving these substantial ploidy shifts.

ALTERNATIVE: Mitosis is an energetically demanding process, requiring substantial ATP and NADH to support replicational, structural and morphological changes, including DNA synthesis, cytoskeleton reorganization, redistribution of membrane-bound organelles, nuclear envelope breakdown and reformation, and chromosome alignment and separation. Despite the overall reduction in anabolic metabolism during mitosis, the energy demand remains considerably high [Skog et al, Exp Cell Res, 1982; Salazar-Roa, Trends Cell Bio, 2017]. To meet this demand, cancer cells employ multiple strategies to enhance ATP production, including upregulation of oxidative phosphorylation (OXPHOS) and glycolysis. Cyclin-dependent kinases (CDK) has been shown to stimulate mitochondrial ATP generation by phosphorylating key components of the respiratory chain complex I in cancer cells [Wang et al, Dev Cell, 2014] and the mitochondrial outer membrane protein TOM6 in yeast [Harbauer et al, Science, 2014]. Additionally, AMPK-mediated phosphorylation of MCU has been found to trigger Ca^{2+} flux into mitochondria, rapidly boosting OXPHOS activity to facilitate mitotic progression [Zhao, Nat. Cell Bio, 2019]. Alongside mitochondrial respiration, glycolysis and lactate production are also upregulated during mitosis, further contributing to ATP supply [Skog et al, Exp Cell Res, 1982]. Aneuploid and whole-genome doubled (WGD+) cancer cells face unique metabolic challenges due to their altered genomic content (reference). These cells require varying levels of key metabolites including glucose, phosphate, and oxygen to sustain mitotic progression and manage the increased biosynthetic burden associated with additional chromosomes (reference). Elevated glucose uptake supports glycolysis and biomass production, while increased phosphate availability is essential for nucleotide synthesis and ATP generation. Oxygen demand is also enhanced, as aneuploid cells rely more heavily on OXPHOS to compensate for energy imbalances (reference). The metabolic rewiring in these cells influences their ability to withstand energy-dependent replicative and mitotic stress, impacting their dependence on specific energy production and metabolic pathways. Despite growing insights into mitotic energy regulation, the mechanisms by which different cancer cell types, particularly those with aneuploidy and WGD, adapt to nutrient limitations during

mitosis remain poorly understood. Cancer cell heterogeneity manifests through genetic, epigenetic, and karyotypic variations. Genetic heterogeneity is well established as a contributor to tumorigenesis, often associated with mutations in genes involved in DNA metabolism (Burrell et al., 2013). Karyotypic variation, encompassing whole-chromosome alterations and ploidy changes, is a widespread phenomenon in cancer cells (Gerlinger et al., 2012) and has been recognized as a hallmark trait of tumorigenesis (Storchova and Pellman, 2004; Weaver and Cleveland, 2006; Hanahan and Weinberg, 2011). This suggests that karyotypic alterations may actively drive different stages of cancer evolution. Supporting this idea, specific aneuploid chromosomes have been implicated in increasing genomic instability (Zhu et al., 2012; Nicholson et al., 2015), enhancing drug resistance (Lee et al., 2011), and promoting tumor growth and oncogenic potential (Ben-David et al., 2014; Rutledge et al., 2016). Karyotypic alterations are thought to fuel cancer progression by tumor relapse and enhanced cellular fitness (Sotillo et al., 2010; Sheltzer et al., 2017). Furthermore, polyploidy and whole-genome doubling have been implicated in tumorigenesis (Fujiwara et al., 2005), drug resistance (Kuznetsova et al., 2015), and genomic instability (Ganem et al., 2009; Silkworth et al., 2009; Kuznetsova et al., 2015). Given its strong association with poor prognosis in cancer patients (Swanton et al., 2009; Danielsen et al., 2016), understanding the fitness advantages of underlying karyotypic heterogeneity remains a critical challenge. Tumor microenvironment is inherently stressful and exposes cancer cells to various unfavorable conditions. These include nutrient deprivation, hypoxia, acidic pH (Fukumura and Jain, 2007), immune surveillance (Dunn et al., 2004), and hyperthermia (Wust et al., 2002; Jha et al., 2016). Such environmental stressors have been implicated in promoting karyotypic alterations, with studies linking chromosomal instability to exposure to DNA damage (Bakhoun et al., 2014), oxidative stress (Gentric et al., 2015), viral infections (Machida et al., 2009), and mitotic slippage following prolonged mitotic arrest (Rieder and Maiato, 2004). Karyotypic instability is known to result from circumstances that increase mitotic defects (Bakhoun et al., 2009; Ganem et al., 2009; Silkworth et al., 2009) or prevent their correction (Cahill et al., 1998; Sotillo et al., 2010; Targa and Rancati, 2018). Alteration of the spindle assembly checkpoint (SAC) signaling cascade, an evolutionarily conserved mechanism that delays anaphase onset by preventing the activation of the anaphase-promoting complex/cyclosome (APC/C) and the degradation of cyclin B1 (Musacchio, 2015) in response to spindle defects, is linked to chromosome instability (Cahill et al., 1998; Hanks et al., 2006; Giam and Rancati, 2015). However, genetic mutations of SAC components other than BubR1 are uncommon in cancer cells (Kops et al., 2005; Solimini et al., 2012), indicating that other nongenetic mechanisms could promote karyotypic heterogeneity. Notably, nutrient deprivation, a common feature of the tumor microenvironment due to inadequate vascularization, may further exacerbate chromosomal instability by impairing mitotic processes and cellular homeostasis. In agreement with the idea that environmental stress could lead to karyotypic heterogeneity, unicellular eukaryotes were shown to undergo aneuploidization and loss of heterozygosity in response to stress or antifungal drug treatment (Forche et al., 2011; Chen et al., 2012a), suggesting that similar mechanisms could lead to karyotype changes or ploidy shifts in cancer cells under metabolic and microenvironmental stress. In this study, we investigated whether tumor microenvironmental stresses can drive karyotypic changes, serving as a mechanism for karyotypic heterogeneity independent of mutations in gatekeeper genes. Our findings revealed that cancer cells underwent karyotypic selection or alterations in response to cellular stress. At the molecular level, we demonstrated that glucose deprivation disrupts telomerase activity (?), triggering mitotic exit without proper chromosome segregation, ultimately leading to genome doubling. Additionally, we found that phosphate starvation selects for a specific karyotype that downregulates tryptophan metabolism by sacrificing de novo NADH synthesis while upregulating the salvage pathway for sufficient nucleotide biosynthesis. These results suggest that environmental stressors can act as a nongenetic driver of genetic and phenotypic variation in tumor cells. Furthermore, they indicate that stress-specific karyotypic adaptations may serve as potential biomarkers for therapies targeting cancer metabolism.

3 Results

We used three cell lines with distinct base line ploidies (MDA-MB-231: ploidy of 3.8; HGC-27: ploidy of 3.3; SNU-668: ploidy of 2.8). We conducted evolution experiments with all three cell lines in standard medium (SM), Phosphate depleted medium (Pdep: MDM-MB-231 only), and Glucose depleted medium (Gdep: HGC-27 and SNU-668 only). To initiate each evolution experiment, we cultured all lines for more than 6 months in each environment with the time of transfer occurring after cells reach to 5 million per flask, which is equal to 80% confluence. At the onset of experiments, we began culturing cells in hypophysiological concentrations of glucose (hypoglycemia: 3 mM) (Cline et al, NEJM, 1999) or phosphate (hypophosphatemia: 0.32 mM) (Bergwitz and Juppner, Adv. Chronic Kidney Dis., 2011). Samples from parental and evolved lineages were frozen after more than 50% decrease was observed in growth rates. We monitored the growth rate of cells while growing under these nutrient deprivations of various durations and intensities. This monitoring informed us when to further decrease Phosphate/Glucose in their media vs. when we should keep the concentration steady so as in to allow the cells to adapt. We evolved a total of 12 lines (1 parental * 3 growth conditions * 2 cell lines * 2 replicates) for more than six months as of now (a minimum of 6 transfers * 10 generations between each transfer).

Karyotyping the cells at different passages during their adaptations to the respective resource deprivation revealed two strong shifts in ploidy: ploidy reduction in MDA-MB-231 following phosphate deprivation (Figure 2 A,E) and ploidy increase in HGC-27 following Glucose deprivation (Figure 2B,F). Additional significant shifts in ploidy were observed in SNU-668 following Glucose deprivation (Figure 2C,D,G,H), but the effect size was small, indicative of karyotype- rather than ploidy

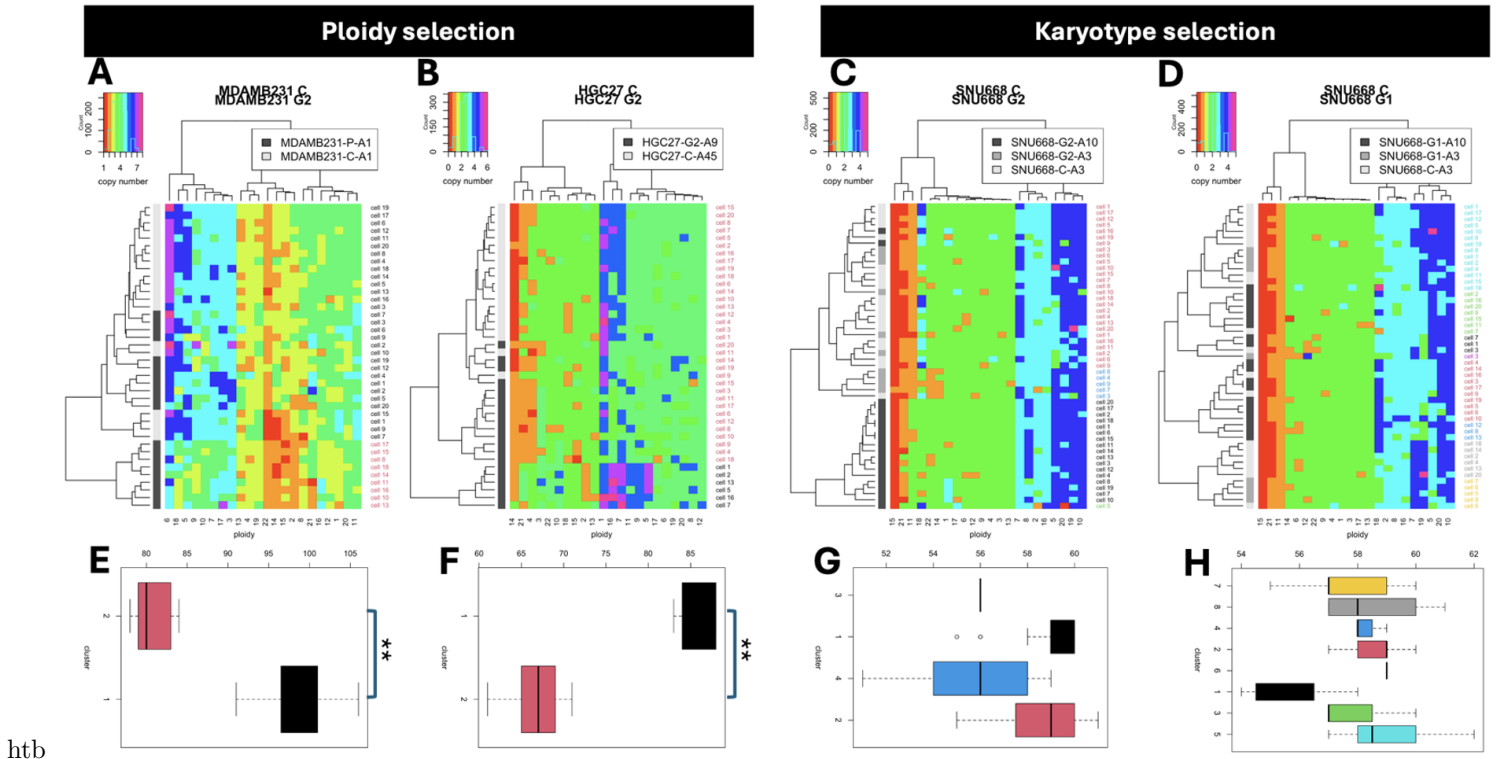


Figure 2: Karyotype evolution facilitates adaptation to low-Glucose and low-Phosphate conditions. (A-D) Cells from control (white) and test samples (gray: early passages, black: late passages) from HGC-27, SNU-668 and MDA-MB-231 were karyotyped during their adaptation to low-Glucose or low-Phosphate conditions. Clustering cells from both test and control samples (rows) according to their chromosome level copy number (columns) shows that cells cluster according to their sample origin in 3 out of 4 lineages (A-C), indicative of directed karyotype evolution in these lineages. Cluster membership of each cell shown as color-coded row names on the right of each heatmap. (E-F) Comparing ploidy across cell clusters for each sample. Ploidy was defined as the total number of chromosome copies per cell. ** indicates p-value of a t-test below 0.005.

selection. Indeed, evidence for directed karyotype evolution (defined as enrichment of late passages among karyotype subtrees) were present in 3 out of 4 lineages (Figure 2).

3.1 Adaptations to phosphate reduction

XX: describe karyotype changes in response to Phosphate deprivation in HGC-27 and SNU-668.

- **mass-spec results** are consistent with cells rewiring metabolism to favor salvage pathway.
- karyotype and/or scSeq results

3.2 Adaptations to glucose reduction

XX: describe karyotype changes in response to Glucose deprivation in HGC-27 and SNU-668.

3.3 Adaptations to oxygen reduction

XX: describe karyotype changes in response to Oxygen deprivation in HGC-27 and SNU-668.

To begin assessing whether this phenomenon is generalizable to other cancer types we repeated the experiments for an isogenic pair of near-2N and near-4N breast cancer cell lineages ...

We also injected 2N and 4N cells in mice to evaluate how their karyotype/ploidy re-adapts to in-vivo conditions...

3.4 Influence of space on karyotype evolution

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4 Methods

The cytokinesis-block micronucleus (CBMN) assay

The CBMN assay is a standard method for measuring chromosome instability (Fenech, Nat Protocols, 2007). Cells were cultured in medium containing 4.5 µg/ml cytochalasin B (Vendor) to block cytokinesis for 24 hours, trypsinized, spun onto slides using a cytospin cytocentrifuge (Vendor), air-dried, fixed in 31 methanol/glacial acetic acid and stained with Wright and Giemsa. Binucleated (BN) cells, micronuclei (MNI), nucleoplasmic bridges (NPBs) and nuclear blebs (NBs) were scored as described previously [Camps et al, FASEB, 2005]. MNI were morphologically identical, their diameters vary between 1/16 and 1/3 of the mean diameter of the main nuclei, not linked to the main nuclei. NPBs were continuous nucleoplasmic bridges between the two nuclei in a BN cell, which is no wider than 1/4 of the nuclear diameter. NBs were small protrusions of the nuclear material, connected to the main nucleus by a thin chromatin segment. For each sample, a total of 1000 BN cells were evaluated for the frequency of MNI, NPBs and NBs. A total of 500 living interphase cells were used for assessment of mono-, bi-, and multi-nucleated cells without adding cytochalasin B.

Fluorescence In Situ Hybridization (FISH)

FISH was performed using pan-centromeric probes which recognize all chromosomes (Vendor). Chromosome-specific centromeric probes (FITC -conjugated alpha-satellite DNA probes for chromosomes 7 and TRITC -conjugated probes for chromosomes 17) were from (Vendor)). Molecular hybridization and immunofluorescent detection were carried out according to the manufacturer's instructions. Fluorescence was observed under Zeiss LSM 710 laser confocal microscope equipped with Zen software to process the image.

Immunofluorescent Confocal Laser Microscopy

Cells on coverslip were fixed with 4% paraformaldehyde in PBS, permeabilized with 0.1% Triton X-100 in PBS, and blocked with 2% bovine serum albumin in sodium phosphate buffer (pH 7.4). Cells were co-stained with anti-pericentrin (1500 dilution of ab4448, Abcam, UK) and anti-alpha tubulin (1500 dilution, ab7291, Abcam, UK) or anti-H2AX(11000 Cell Signaling) and anti-53BP1(11000 dilution, 4937s, Cell signalling, USA) according to the protocols of the suppliers. Alexa Fluor 633-conjugated goat anti-mouse and Alexa Fluor 488-conjugated goat anti-rabbit antibody (1500) were from Molecular Probes (Invitrogen, USA). Nuclei were counterstained with DAPI. Fluorescence was observed under Zeiss LSM 710 laser confocal microscope equipped with Zen software to process the image.

Cell cycle assay and immunofluorescent detection of phosphorylated histone H3

1×10⁶ cells were collected and fixed in 70% ethanol (4°C, 48 hours). After fixation, the cells were washed with PBS, and permeabilized with 0.3% Triton X-100 in PBS for 15 min. Then the cells were blocked in 1% BSA for 10 min and stained with antibody that specifically recognizes the phosphorylated form of histone H3 (11000 dilution, 3377s, Cell signalling, USA). After incubated for 1 hr in room temperature, the cells were rinsed with PBS and incubated with FITC-conjugated IgG antibody for 30 min. Then the cells were rinsed with PBS twice, and stained with the CycleTEST™ PLUS DNA Reagent Kit (BD Biosciences, USA). The samples were analyzed by the FACScan™ Calibur (Becton Dickinson, USA). The FCS Express version 3 (De Novo Software) was utilized to analyze the data.

Western blot analysis

Cells were lysed with RIPA Buffer, supplemented with protease inhibitor cocktail (BioRad, USA). Protein concentration was measured by Bradford protein assay (BioRad, USA). After heat denaturation, samples were stored at 80°C before use. The protein was applied to a 4% to 15% SDS polyacrylamide gel, transferred to a PVDF membrane, and then detected by the proper primary and secondary antibodies before visualization by Western Lighting Plus ECL kit (Perkin Elmer, USA). The primary antibodies used: rabbit anti-mad2, rabbit anti-bub1b, rabbit anti-bub3 and rabbit anti-cdc2 (Cell Signalling

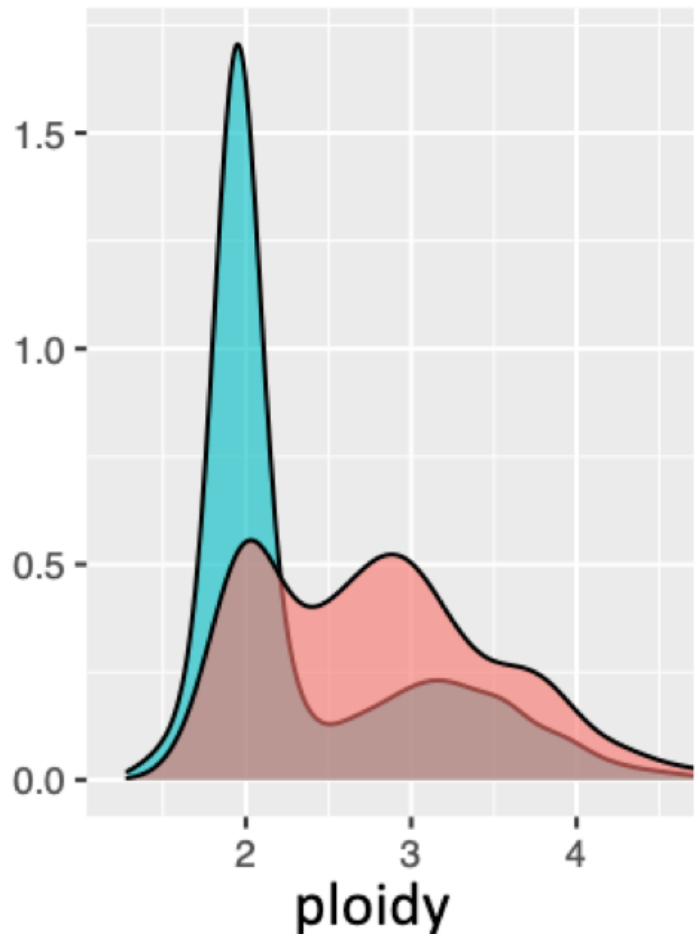


Figure 1: .

Technology, USA), rabbit anti-cyclin B1 and mouse anti-bub1 (Cell signaling, USA).

Time-lapse imaging of mitotic progression MDA-MB-231 (GFP-H2B) cells were seeded on an 8-chambered cover glass (μ -Slide 8 Well, ibidi; 80826). For live cell imaging, the original culture medium was replaced by RPMI (Glucose, 1.5 mM, GIBCO, Waltham, MA, USA) or complete RPMI media (Gibco, Grand Island, NY, USA; A1443001). Images were collected every 3 min with an exposure time of 0.2 second for 10–20 hours, using a 20 $\times$ objective on an inverted fluorescence microscope (Nikon Eclipse Ti-E) or Delta Vision microscope (GE Healthcare, Buckinghamshire, UK). The temperature of the imaging medium was kept at 37°C. Image sequences were viewed and analyzed with Image J.

MDA-MB-231 (GFP-Bub3) stable cells were seeded on the glass-bottom cell culture dish (NEST; 801001). Images were acquired on a Delta Vision microscope with a 60 $\times$ objective at 4 minutes intervals, with optical sections acquired at 1 μ m apart in the z-axis. The final data were shown as maximal intensity projections. The temperature of the imaging medium was kept at 37°C. Image sequences were viewed and analyzed with Image J.

Immunofluorescence and inter-kinetochore distance analysis

For immunofluorescence analyses, cells were plated on chamber slides. Cells were fixed with 4% paraformaldehyde for 10 minutes. Next, the cells were incubated with 0.2% Triton X-100 in PHEM for 10 minutes and then blocked with 1% BSA in TBST for at least 30 minutes. The primary antibody was incubated 2 hours at room temperature followed by secondary antibody incubation and DAPI staining. Finally, the cells were mounted using DAKO Fluorescence Mounting Medium (Dako North America, Inc, Carpinteria, CA, USA; S3023). Images were acquired using a DeltaVision Deconvolution microscope (GE Healthcare, Buckinghamshire, UK).

For microtubule (MT) stability assay, cold treatment was performed by incubating cells in cold media (4°C) on ice for 10 min before fixed (Yang et al, Mol Cell Bio, 2005).

To determine the inter-kinetochore distance, cells were fixed with 4% paraformaldehyde and processed as described above. The center-to-center distances between sister-kinetochores were measured from different stacks. When sister-kinetochores were in the same focal plane, the inter-kinetochore distance was measured as previously described (Gao et al, Nature Cell Bio, 2011). Images were collected with a DeltaVision Deconvolution microscope and analyzed with Image J.

Metaphase spreads

Cells were harvested by trypsinization and washed once in phosphate-buffered saline (PBS). Cells were resuspended gently in 100 μ l of PBS, and 75 mM potassium chloride (Kanto Chemical, Tokyo, Japan) was added slowly up to a volume of 10 ml. The samples were then incubated for 10 min in a 37°C water bath. Following that, 1 ml of Carnoy solution (3:1 methanol:acetic acid) was added to each sample, and samples were centrifuged for 10 min at 1000 rpm. Supernatant was discarded and cells were gently resuspended in a small amount of the remaining supernatant. Carnoy solution (10 ml) was added slowly to the samples and cells were fixed for at least 1 h at room temperature (RT). Samples were centrifuged for 10 min at 1000 rpm, supernatant was discarded, and cells were fixed again with 10 ml of fresh Carnoy for at least 1 h at RT. Samples were then centrifuged and pellets were resuspended in a suitable volume (0.5–1.5 ml) of Carnoy solution. A sample of 20 μ l of this suspension was drawn up and dropped from a height (10–30 cm) onto glass slides and allowed to air-dry. DNA was then stained with Giemsa (Life Technologies, Thermo Fisher Scientific).

Metaphase spreads either were visualized with AxioImager Z1 (Carl Zeiss, Oberkochen, Germany) and images taken with either an AxioCam MRm (Carl Zeiss) or a CoolSNAP HQ2 CCD camera (Photometrics, Tucson, AZ), or were visualized with DM6000 B (Leica, Wetzlar, Germany) and images taken with CV-M2CL (JAI, Yokohama, Japan), or were visualized with Imager.Z2 (Carl Zeiss) and images taken with CoolCube 1 (MetaSystems). For each time interval and condition, images were captured from a field of view with clear 4,6-diamidino-2-phenylindole (DAPI) (Sigma-Aldrich) staining and the cells were analyzed sequentially subsequently, starting from the first captured image, with noise manually excluded, until the specified number of cells to minimize bias was achieved.

References

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